
Campus Parking and Exit Carpool Coordinator: Analysis, Design and Simulation-Based Validation of a Sociotechnical System

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Abstract— Sustainable mobility within university campuses is a critical factor in mitigating parking deficits, traffic congestion, and localised pollution. This paper reports the comprehensive lifecycle of analysis, design, and simulation-based validation of the Campus Parking and Exit Carpool Coordinator, a system tailored for the Faculty of Engineering at the Universidad Distrital Francisco José de Caldas (Sabio Caldas and Administrative campuses). Based on field observations, occupancy metrics, and a survey conducted with 51 users, a daily overflow of approximately 90 motorcycles onto public roads was identified, negatively impacting the surrounding area, alongside a predominant reliance on informal coordination via messaging platforms. A layered architecture is proposed, integrating parking inventory management, policy-based reservations, and carpooling ride-matching. Validation through process-oriented and agent-based behavioural simulation, executed across three contrasting scenarios (baseline, optimisation, and failure), confirms the design viability and quantifies critical risks: data synchronisation rates and cancellation management must be treated as a joint risk to prevent trust collapses. The results position the system for a future controlled implementation.

Keywords—Carpooling, sustainable mobility, university parking, event simulation, agent-based modelling, systems engineering

1 Introduction

Urban university campuses frequently face mobility deficits driven by the predominance of private vehicles, which saturates available parking spaces, generates queues at access points, and displaces the overflow onto surrounding public roads [1]. At the Faculty of Engineering of Universidad Distrital Francisco José de Caldas (UDFJC), with campuses on Calle 40 (Sabio Caldas) and Calle 34, this problem is intensified by limited internal infrastructure designed for cars and motorcycles, poor management, and the systematic occupation of public space by approximately 90 motorcycles daily in the surroundings of the Calle 40 campus. This configuration not only restricts the effective capacity of the campus, but also generates conflicts and safety risks for users and neighbours, and contributes to a negative environmental impact associated with short urban trips that could be served by more sustainable modes [2].

Studies on parking and sustainable mobility in Bogotá indicate that policies oriented towards providing abundant and low-cost parking produce structural inefficiencies and encourage excessive car use [1]. Complementary research on Latin American university campuses shows that coordinated interventions such as carpooling schemes and bicycle promotion can reduce local pollutants and greenhouse gas emissions, while local policy documents such as the Bogotá Sustainable Mobility Plan [3] stress the need for closer alignment between institutional transport management and city-wide sustainability goals.

A systems analysis of the engineering campuses reveals marked temporal and spatial imbalances: arrivals are heavily concentrated in the morning (55 % of users arrive before 08:00), internal parking is scarce or absent at some sites, and motorcycle overflow is displaced onto the public roads surrounding the Calle 40 campus. Field observations, parking measurements and process documentation show that vehicle access control is carried out through a combination of a digital registration application and manual decisions by security staff, which generates bottlenecks at the basement entrances.

Responses from 51 participants in a structured survey reveal a strong dependence on public transport, targeted use of private vehicles for certain trips, and informal ride-sharing arrangements coordinated mainly through messaging applications, motivated by cost reduction and improved perceived safety. At the same time, respondents report a high willingness to adopt an institutional shared transport system and a relevant level of interest in a formalised carpooling alternative.

These empirical findings motivate the Campus Parking and Exit Carpool Coordinator: a system that integrates parking inventory management, policy-based reservations, and ride-matching within a verified institutional environment. This paper describes the complete development cycle of the project, from systems analysis (Workshop 1) through conceptual architecture (Workshop 2), robust design and project management (Workshop 3), and simulation-based validation

(Workshop 4). The following sections describe the methods employed, present the main patterns and constraints of the current mobility system, detail the proposed architecture and risk framework, and report the simulation results that validate the design prior to a pilot deployment.

2 Methods and Materials

2.1 Systems analysis approach

This study applies a systems analysis approach to the transportation and parking system of the UDFJC engineering campuses. The objective is to identify elements, interactions, and bottlenecks that affect mobility, safety, and environmental performance, providing an empirical basis for the concept of shared mobility coordination.

2.2 Data collection

Field observations and measurements were carried out at the campus (Calle 40), characterising the available infrastructure and typical use, including the motorcycle-oriented basement in the building and the presence of approximately 90 motorcycles parked on public roads near the main entrance. In parallel, a structured survey of 51 community members captured user profiles, main transport modes, arrival times, parking difficulties, and willingness to adopt formal carpooling. Process documentation described how vehicle access control is managed through a digital registration application and manual decisions by security staff, and how informal carpooling is coordinated through messaging applications.

2.3 Simulation design

Two complementary models were developed for system validation. The first is a *Process-Oriented Simulation* based on event simulation, which captures the life cycle of parking spaces and the exit carpooling matching process. The second is a *Behaviour-Oriented Simulation* based on an agent-based model with 500 simulated students over 60 academic days, which evaluates platform adoption, user trust, and parking saturation through four feedback connections.

3 Results and Discussion

3.1 Analysis of the current system

Systems analysis reveals a critical misalignment between the institutional demand for mobility and the available infrastructure. The finding that 55 % of users arrive before 08:00 generates a synchronised peak load that overflows the basement capacity at the Calle 40 campus and manifests in the observed overflow of approximately 90 motorcycles onto public roads near the main entrance. This pattern confirms that fluctuations in demand during the morning window can trigger queues at access points and the displacement of parking toward the surrounding public space.

3.2 Survey results

Survey results reveal that, although the majority of respondents rely on public transport for their daily commute, arrivals are heavily concentrated before 08:00, coinciding with periods of peak saturation in the basement and surrounding streets. The non-negligible interest in formal carpooling indicates behavioural readiness for shared and low-emission modes, although these options do not yet have coordinated institutional tools.

3.3 Proposed coordination concept

The layered coordination concept of the Campus Parking and Exit Carpool Coordinator proposes a modular architecture that connects the elements and relationships identified in the systems analysis. The client layer hosts a mobile/web application for students and a dashboard providing access to ride-matching, notifications, and status information. The application layer groups the core services for ride-matching, messaging, and manual status updates. At the data layer, user and reputation records, as well as trip and parking information, are stored. Finally, a monitoring layer consumes these data to generate system status indicators and alerts for operators.

3.4 Process-oriented simulation results

In the baseline scenario (S1), daily motorcycle demand fluctuates according to ranges by day of the week, with Friday consistently generating the highest demand (180-240 motorcycles). Given the capacity of 110 spaces, overflow events occur regularly on high-demand days. With a match rate of 45 %, approximately half of carpooling requests are successfully coordinated, while synchronization failures (15 %) and cancellations (10 %) reduce the overall reliability of the system. Rain events, with a probability of 35 %, amplify parking demand by a factor of 1.25, pushing overflow to its maximum boundary of 90 vehicles on affected days. In the optimization scenario (S2), the improved configuration produces substantial improvements across all metrics. The increase in match rate to 75 % nearly doubles successful carpooling coordination compared to the baseline scenario, while the synchronization failure rate drops to 3 % and the cancellation rate falls to 4 %. The combined effect of higher matching efficiency and lower disruption rates results in significantly reduced overflow. In the failure scenario (S3), the system exhibits the worst performance across all metrics. The match rate collapses to 30 %, synchronization failures spike to 40 %, and cancellations reach 25 %. Overflow vehicles consistently reach the maximum boundary of 90 units, confirming the cascading effect of infrastructure degradation on user behavior and parking saturation simultaneously.

3.5 Behavior-oriented simulation results

The behavior-oriented simulation was executed over 60 operational days, tracking the evolution of platform adoption, user trust, and parking saturation through four active feedback loops. In S1, without institutional incentives,

platform adoption remains stable near its initial value of 30 %, with no significant growth stimulus. The trust score fluctuates around its initial value of 50 %, driven by the balance between successful matches (which generate trust gains of +0.015 per match) and cancellations and synchronization failures. WhatsApp usage remains dominant, reflecting the system's inability to displace informal coordination without external incentives. In S2, the adoption bonus of 10 % per cycle produces consistent growth in platform adoption over the 60-day period, driving adoption toward its upper boundary of 40 %. As adoption increases, more drivers become available, successful matches increase, and trust compounds upward. The system converges toward a high-trust, high-adoption equilibrium by the end of the simulation period. In S3, the adoption penalty of -30 % drives adoption toward its minimum boundary of 15 % rapidly. Compounding synchronization failures (40 %) trigger repeated trust losses, while the high cancellation rate (25 %) further erodes trust. The system converges toward a low-trust, low-adoption state where WhatsApp usage dominates and parking saturation remains consistently high.

3.6 Emergent behaviour

The behaviour-oriented simulation reveals two emergent phenomena not explicitly programmed as direct outcomes.

Adoption-saturation feedback: when parking saturation exceeds 90 %, the adoption feedback loop activates, generating upward pressure on platform usage. This creates a reinforcing cycle in S2 where high demand drives adoption, which drives carpooling coordination, which reduces demand, generating a stabilising negative feedback loop not observable in the process-oriented model in isolation.

Trust collapse under compounding failures: in S3, the simultaneous activation of the cancellation and synchronisation-failure feedback loops produces a trust degradation rate that exceeds that of any isolated failure mode. Once trust falls sufficiently, the match rate decreases (since trust directly scales the matching probability), which reduces successful matches, which eliminates the trust recovery mechanism. The system enters a self-reinforcing degradation cycle with no internal recovery path once trust approaches zero.

3.7 Design validation

The simulation results validate or challenge the following key design decisions from previous workshops. The 110-space basement capacity as the operational boundary is *validated*: overflow occurs consistently on high-demand days, confirming that capacity is insufficient without carpooling coordination. The carpooling matching strategy as an overflow mitigation measure is *validated*: higher match rates (75 % in S2 versus 45 % in S1) directly reduce motorcycle arrivals and overflow events. Infrastructure reliability as a trust dependency is *validated*: a synchronisation failure rate of 40 % in S3 triggers a cascading trust collapse.

On the other hand, platform adoption as self-sustaining above a threshold is *partially challenged*: adoption

boundary behaviour reveals a structural ceiling at 40 % under current driver-ratio and vehicle-capacity constraints, suggesting these parameters may need revision to enable growth beyond the current ceiling. Similarly, the treatment of rain events as isolated demand spikes is *challenged*: rain probability interacts with day-of-week demand to create compounded high-risk windows specifically on rainy Fridays, requiring a dedicated response protocol beyond standard daily management.

4 Risk Management and Quality

The system design must consider both technical and human dimensions. Its robustness depends on the interaction between parking rules, matching logic, operational control, institutional policy, and user behaviour.

4.1 Critical risks identified

The main risks identified and quantified by the simulation are as follows.

R1 — Low platform adoption: without active institutional incentives, adoption stabilises near the initial 30 % with no growth trajectory, independent of system performance. The simulation confirms that the adoption bonus is the only parameter capable of driving adoption toward its upper boundary.

R2 — Data desynchronisation: a synchronisation failure rate of 40 % in S3 triggers a cascading trust collapse through compounding feedback loops. This risk carries the most destructive compounding effect on trust and adoption simultaneously.

R3 — Cancellation and no-show behaviour: the simultaneous activation of cancellation effects and synchronisation failures produces a trust degradation rate that exceeds that of any isolated failure mode. For this reason, infrastructure reliability and cancellation management must be treated as a joint risk, not two independent failure modes, when designing mitigation strategies.

R4 — Peak-demand overload: the interaction between rain probability (35 %) and Friday demand (180–240 motorcycles) creates a compounded high-risk window where effective demand can reach 225–300 motorcycles, consistently exceeding the 110-space capacity and pushing overflow to its maximum boundary of 90 vehicles.

4.2 Quality attributes (ISO/IEC 25010)

The system quality framework aligns ISO/IEC 25010 attributes with simulation findings. *Performance efficiency* requires the system to handle up to 500 simultaneous requests during the 07:00–08:00 and 12:00–14:00 windows, the peak saturation periods identified in the survey. *Reliability* demands that the synchronisation failure rate remain below 5 % under normal operation, given that values above 15 % generate significant trust fluctuation. *Usability* requires that publishing a carpooling slot take fewer than three interactions, in response to the finding that only 34.3 % of respondents expressed willingness to use

carpooling, making minimal interface friction critical. *Security* demands role-based access control and traceability of all relevant operational events.

5 Design Recommendations

Based on the simulation findings, the following modifications are recommended for the system architecture.

Rainy Friday response protocol: the interaction between rain probability and Friday peak demand creates the highest-risk overflow condition in the system. A dedicated operational protocol should be established for days where both conditions are predicted simultaneously, including proactive overflow redirection to external parking zones and early user notifications to encourage carpooling coordination before demand peaks.

Trust recovery mechanism: the simulation reveals that once trust collapses under compounding synchronisation failures and cancellations, no internal recovery path exists within the current model. A dedicated trust recovery feature should be incorporated into the architecture, such as service credits, incident acknowledgement notifications, or temporary incentive boosts following detected failure events.

Adoption ceiling investigation: the current structural constraints (15 % driver ratio, average capacity of 3 passengers) produce a hard adoption ceiling at 40 %. A design review of the driver incentive structure is recommended to evaluate whether increasing the driver ratio or offering capacity incentives for larger vehicles could raise this ceiling.

Institutional incentive strategy: the institutional incentive should be active from the first day of deployment and maintained continuously rather than applied as a temporary campaign. Given the adoption ceiling at 40 % under current structural constraints, the incentive strategy alone is insufficient to generate a self-sustaining network effect; it must be accompanied by the driver-ratio improvements noted above.

6 Conclusions

This work presents the complete systems engineering lifecycle for the Campus Parking and Exit Carpool Coordinator, from analysis (W1) through design (W2), management and risk (W3), to empirical simulation-based validation (W4). The principal conclusions are as follows.

Design validation confirms that the core architectural decisions from Workshop 2 are supported by simulation evidence. The 110-space basement capacity is confirmed as structurally insufficient without active carpooling coordination, validating the platform's role as a necessary operational component. The match rate parameter demonstrates a direct relationship with overflow reduction: the optimisation scenario (75 %) produces substantially lower overflow than the baseline (45 %), confirming that matching efficiency is the primary lever for parking management.

Risk quantification converts the qualitative concerns of Workshop 3 into observable simulation outcomes. Low adoption without institutional incentives produces stable but stagnant engagement near the 30 % baseline, while the optimisation scenario drives adoption toward its structural ceiling of 40 %, validating the incentive strategy as a necessary and effective mitigation. Infrastructure reliability is confirmed as the highest-impact risk: a synchronisation failure rate of 40 % in the failure scenario triggers a compounding trust collapse that no other single parameter can produce.

Emergent discoveries reveal behaviours not visible in the conceptual design phase: the compounding interaction between synchronisation failures and cancellations produces trust degradation at a greater rate than either failure mode in isolation; the interaction between rain probability and Friday peak demand creates a compounded high-risk overflow condition requiring a dedicated operational protocol; and the adoption ceiling at 40 % under current structural constraints reveals a growth limitation that incentive strength alone cannot overcome.

Implementation readiness positions the system for a credible pilot deployment. The key operational parameters established through simulation — the 110-space capacity boundary, the 30 % initial adoption baseline, the critical role of infrastructure reliability, and the necessity of continuous institutional incentives — provide a concrete foundation for Phase 1 deployment planning.

This work demonstrates that simulation-based validation is not a final step but a feedback mechanism that iteratively strengthens design decisions and surfaces gaps that formal analysis cannot reveal. The models developed here will serve as living benchmarks: as real deployment data becomes available, the simulations can be recalibrated and used to evaluate future design modifications before they are committed to production.

7 Acknowledgements

The authors thank Engineer Carlos Andrés Sierra, M.Sc., lecturer of the Systems Analysis and Design course at Universidad Distrital Francisco José de Caldas, for his methodological guidance throughout the development of this project. Gratitude is also extended to the 51 survey participants from the Faculty of Engineering for their collaboration in data collection.

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